

Expert Goggles: Detecting and Annotating Visualizations using a Machine Learning Classifier

Stephen MacNeil
Temple University
Philadelphia, PA, USA
stephen.macneil@temple.edu

Parth Patel
Temple University
Philadelphia, PA, USA
parth.patel02@temple.edu

Benjamin Smolin
Temple University
Philadelphia, PA, USA
ben.smolin@temple.edu

ABSTRACT

Data visualizations are an increasingly common way to communicate information online. However, not everyone has the data literacy skills necessary to interpret complex visualizations effectively. We present Expert Goggles, a Chrome browser extension that provides just-in-time guidance to help non-experts interpret the visualizations they encounter on the web. Expert Goggles uses a machine learning classifier to automatically determine the visualization type and uses this context to deliver relevant learning materials. We discuss how this approach to automatically detect the context and provide just-in-time support might transform everyday experiences into informal learning opportunities.

ACM Reference Format:

Stephen MacNeil, Parth Patel, and Benjamin Smolin. 2022. Expert Goggles: Detecting and Annotating Visualizations using a Machine Learning Classifier. In *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*, October 29–November 2, 2022, Bend, OR, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3526114.3558627>

1 INTRODUCTION

Data visualizations are becoming more common in news articles, social media, and health communities to communicate important trends and data [12]. Despite a general lack of data literacy skills [5], people often overestimate their ability to interpret data visualizations [16]. This lack of ability combined with overconfidence can result in misinterpretations and lead to misinformation. Additionally, technological and financial barriers lead to lower access to high-quality data literacy training [14]. For those who receive formal training, it is often decontextualized from the contexts in which trainees encounter data and data visualizations in their daily lives. Interfaces that provide automatic guidance could help meet this need for data literacy at scale by creating informal learning opportunities inside a user's typical web browsing experience.

Leveraging context can be a powerful technique to facilitate learning [7, 8, 10, 13, 15]. Existing user interfaces, such as Tutorons [7], VoxLens [15], and ScholarPhi [8] have demonstrated how heuristics can be used to provide just-in-time annotations to help learners. In authentic learning settings, like web browsing,

data visualizations can be implemented in myriad ways; making the context challenging to infer through heuristics or pattern matching.

We introduce Expert Goggles, a browser extension that leverages a machine learning classifier to automatically detect visualizations and provide contextually relevant learning materials. As users encounter visualizations on the web, the context of what type of visualization they are viewing provides an opportunity to provide just-in-time informal learning. By annotating the visualization with guidance for interpretation, Expert Goggles may be able to deliver expertise just when it is needed. Our model is comparable to state-of-the-art approaches for classifying visualizations; however, none of these models will always correctly identify the visualization type. We discuss our future work to evaluate Expert Goggles and to explore what happens when the context is incorrectly determined and how to assist users in those scenarios.

2 RELATED WORK

Previous UIST research has developed interfaces that can provide in-context and contextualized learning opportunities. ReMap explored how multimodal interaction can facilitate in-task help-seeking [6] to bridge the gap between unfamiliar software tools and learning resources. Through a combination of speech commands and mouse clicks, users could search through indexed video tutorials. They inferred relevant context by listening for the keywords 'this' or 'that' and a mouse click on a detectable area, such as the crop button.

Blocks-to-CAD introduced the concept of cross-interface bridges which leverage a user's knowledge of a familiar application to learn the skills required to use an unfamiliar one by gradually adapting the interface [9]. By incrementally introducing new concepts and interface elements in a familiar context, users were better able to transfer the knowledge and skills learned in one user interface to another. While these projects leverage context to teach users to use an interface, interfaces can also leverage context to deliver learning material through context-sensitive annotations. For example, Tutorons detects code snippets and uses annotations to help readers understand them [7]. Tutorons identifies explainable regions by searching for relevant HTML tags and parsing them based on language-specific criteria. In the explainable regions, Tutorons generates an in-place annotation with text and images to explain code snippets. VoxLens, is a JavaScript library that allows developers to pregenerate summaries of a data visualization to provide these contextual summaries to a user on-demand [15].

3 EXPERT GOGGLES

Expert Goggles is a Chrome browser extension that provides just-in-time learning opportunities when users encounter visualizations on the web. When users encounter D3.js (D3) visualizations on

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '22 Adjunct, October 29–November 2, 2022, Bend, OR, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9321-8/22/10.

<https://doi.org/10.1145/3526114.3558627>

the web, a sidebar opens automatically to present the user with contextually relevant help for interpreting that type of visualization. While prior work leverages heuristics to detect opportunities to provide guidance [6, 7, 9, 13], Expert Goggles uses a machine learning (ML) classifier to account for the fact that visualizations can be implemented in many ways.

3.1 Automatically annotating user interfaces

To provide just-in-time annotations to facilitate learning, Expert Goggles displays a button in the top right corner of the web page when a visualization is detected. This button opens an interface containing text-based and visual guidance about how to interpret the detected visualization. Figure 1 shows a visualization on the Pudding data blog¹ with guidance in the sidebar.

Expert Goggles leverages predefined guides from The Data Visualization Catalog [1], an encyclopedia of data visualizations. While the guidance provided could instead be a YouTube video or other media (e.g.: ReMap [6]), this paper focuses on detecting the context in real-world settings. Previous work provided learning opportunities through annotation but the annotations were required to be developed alongside the primary interface. However, detecting the context for providing annotations automatically is our primary contribution. Developing interfaces like Expert Goggles that automatically provide contextually relevant annotations to facilitate learning have the potential to save the user’s time as explanations are just-in-time. Moreover, annotations can serve as an introduction to new topics a user can research later. For example, providing a correct annotation for a Sunburst diagram can help the user do more research on them by providing a foundation of knowledge.

3.2 Detecting the visualization context with feature extraction and classification

Prior work has developed ML models to classify visualizations based on SVG elements in a webpage [3] and images [2]. Instead of using SVG elements as features, Expert Goggles listens for function calls made by D3 to construct a feature vector. Function calls are used because they provide additional information that is not explicitly represented in the SVG elements, such as data transforms and scales.

Expert Goggles listens for 23 function calls across the multiple versions of D3. The resulting feature vector includes: channels (scale types, area, pie, .etc), marks (circles, rectangles, etc.), and discriminating functions that aid in the placement of visualization elements (treemap, stack, axis placement, pie, etc.). These features were chosen because they represent the visual encoding of data [4, 11]. These feature vectors are stored as logical vectors, containing only 1s and 0s. We leveraged our domain knowledge by assigning additional weights to marks (e.g.: circle) and discriminating function calls (e.g.: .treemap()). Marks were given a weight of 3 because they represent how data items are represented. Discriminating function calls were weighted 5 because they are unique to specific visualization types. Channels (e.g.: scales) were retained a weight of 1 as they are not unique to any visualizations.

Expert Goggles uses the k nearest neighbors (KNN) algorithm to classify the visualization type. KNN was chosen due to its interpretability and performance with small datasets. To train the

model, 6 examples from each class were randomly assigned into training or testing. The trained model was then evaluated on the testing data. For cross validation, we repeated this process 10 times and averaged class recall, accuracy, and F1 scores across these ten trials to reduce the effects of outliers.

3.3 Classifier Results

While classification accuracy is not our primary contribution, our results (summarized in Table 1), are comparable with existing state-of-the-art methods (72% accuracy and a weighted F1 score of 0.81) [3]. Our model achieved an accuracy of 80.4% and a weighted F1 score of 0.77. Each class was detected with 67% recall or higher, except scatter plots. Scatter plots share many features with other chart types and are difficult to classify. Some classes had perfect recall which is likely due to the small size of the dataset.

Visualization Type	Average Recall	Num. Examples
Pie Chart	1.0000 $\pm$ 0	7
Scatter Plot	0.2929 $\pm$ 0.10	14
Bar Chart	0.8375 $\pm$ 0.14	8
Line Chart	0.7400 $\pm$ 0.13	10
Stacked Bar Graph	0.8538 $\pm$ 0.08	13
Area Graph	1.000 $\pm$ 0	8
Parallel Coord. Chart	0.6333 $\pm$ 0.12	9
Geo Map	0.7111 $\pm$ 0.10	9
Treemap	0.8571 $\pm$ 0.09	7
Stream Graph	0.8700 $\pm$ 0.05	10
Sunburst Diagram	1.000 $\pm$ 0	10
Box Plot	0.8571 $\pm$ 0.09	7

Table 1: The mean recall, standard deviation, and number of examples for each visualization type. The overall accuracy for our classifier was 80.4% with a weighted F1 score of .77

4 DISCUSSION

We presented Expert Goggles, a system that provides just-in-time context-aware annotations for D3 visualizations on the web. Expert Goggles focuses on the D3 library because it is widely used by across the web and on news websites (New York Times, Washington Post, and Wall Street Journal). Expert Goggles currently provides contextual help in the form of a sidebar on the same web page. However, the low-level control provided by D3 presents future possibilities to provide more specific, in-situ guidance. For example, Expert Goggles could eventually leverage the D3 library to anchor guidance to relevant SVG elements, such as axes, points, and lines. These annotations might be able to leverage the visualization type and additional details. For example, a tooltip can be anchored to Stream Graph’s x-axis to explain that it refers to time.

Expert Goggles currently detects 12 visualizations with 84% accuracy across classes. This performance is on par with state-of-the-art models. However, when using ML to detect relevant contexts to provide learning scaffolds, the classifier will occasionally be wrong. In our case, performance was lowest for scatter plots. In those cases, we plan to eventually provide mitigation strategies to help users overcome these misclassifications. One possible mitigation strategy is to display the top three predicted visualizations and have the

¹<https://pudding.cool/2019/04/eu-regions/>

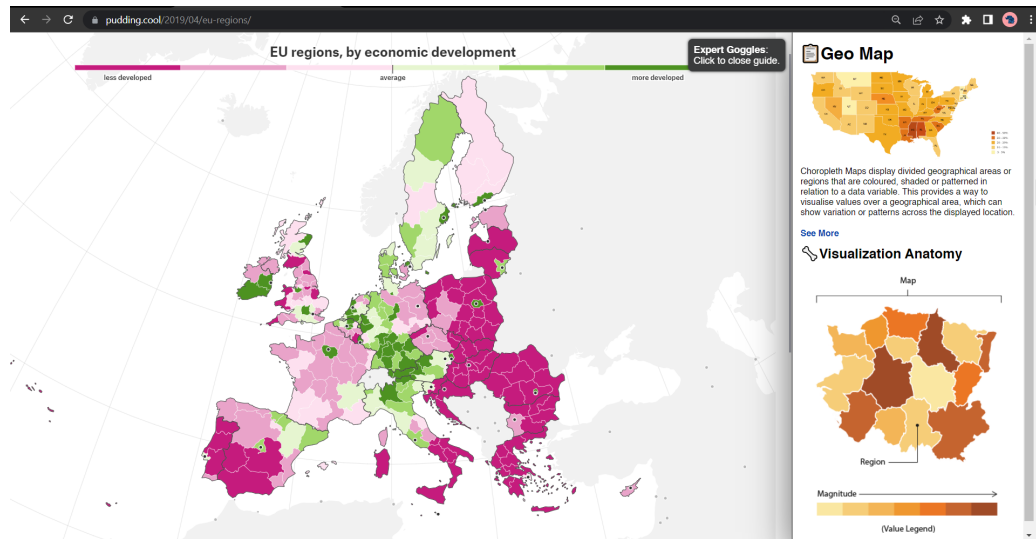


Figure 1: Expert Goggles detects a Geo Map visualization on a news website and opens a sidebar with just-in-time guidance.

user pick which one they are seeing. A second option might be to leverage the confusion matrix to display visualizations that are commonly mistaken for the detected visualization type.

As future work, we will examine the potential to facilitate informal learning and what happens when visualizations are misclassified. Given the imperfect nature of machine learning classifiers and detecting context in real-world settings, it is essential to explore methods to provide help to users when the classifier is wrong. We also hope to expand on the idea of Expert Goggles and generalize the idea further. For example, it is possible to develop augmented reality (AR) annotation systems that provide just-in-time learning materials by detecting context through synchronous video recording and computer vision. As a concrete use case, Expert Goggles might overlay annotations on a car engine or on plants in a garden to guide novices to learn conceptual and procedural skills in authentic learning environments.

5 CONCLUSION

In this paper, we presented Expert Goggles, a tool for automatically detecting and annotating interfaces with just-in-time learning materials. Expert Goggles detects the relevant context, which is a visualization, by analyzing a visited webpage using a machine learning classifier. It then displays contextually relevant guidance for the visualization detected. As future work, we plan to formally evaluate Expert Goggles in user studies and field deployments to understand its capacity to support informal learning. We also plan to investigate the effect of misclassification on user trust in the tool and develop mitigation strategies.

REFERENCES

- [1] 2022. The Data Visualization Catalog. <https://datavizcatalogue.com/>.
- [2] Filip Bajić, Josip Job, and Krešimir Nenadić. 2020. Data Visualization Classification Using Simple Convolutional Neural Network Model. *International journal of electrical and computer engineering systems* 11 (04 2020), 43–51.
- [3] Leilani Battle, Peitong Duan, Zachery Miranda, Dana Mukusheva, Remco Chang, and Michael Stonebraker. 2018. Beagle: Automated Extraction and Interpretation of Visualizations from the Web. (2018), 1–8.
- [4] Jacques Bertin. 1983. *Semiology of graphics; diagrams networks maps*. Technical Report.
- [5] Katy Börner, Adam Maltese, Russell Nelson Balliet, and Joe Heimlich. 2016. Investigating aspects of data visualization literacy using 20 information visualizations and 273 science museum visitors. *Information Visualization* 15, 3 (2016), 198–213.
- [6] C. Ailie Fraser, Julia M. Markel, N. James Basa, Mira Dontcheva, and Scott Klemmer. 2020. *ReMap: Lowering the Barrier to Help-Seeking with Multimodal Search*. Association for Computing Machinery, New York, NY, USA, 979–986.
- [7] Andrew Head, Codanda Appachu, Marti A. Hearst, and Björn Hartmann. 2015. Tutorons: Generating context-relevant, on-demand explanations and demonstrations of online code. In *2015 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. 3–12. <https://doi.org/10.1109/VLHCC.2015.7356972>
- [8] Andrew Head, Kyle Lo, Dongyeop Kang, Raymond Fok, Sam Skjonsberg, Daniel S. Weld, and Marti A. Hearst. 2021. Augmenting Scientific Papers with Just-in-Time, Position-Sensitive Definitions of Terms and Symbols. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (Yokohama, Japan) (CHI '21)*. Association for Computing Machinery, New York, NY, USA, Article 413, 18 pages. <https://doi.org/10.1145/3411764.3445648>
- [9] Ben Lafreniere and Tovi Grossman. 2018. Blocks-to-CAD: A Cross-Application Bridge from Minecraft to 3D Modeling. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology (Berlin, Germany) (UIST '18)*. Association for Computing Machinery, New York, NY, USA, 637–648.
- [10] Stephen MacNeil, Josh Whitka, Aaron Wile, Parth Patel, Emily Jao, and Margaret Hanley. 2022. A Context-Aware Browser Extension for Just-in-Time Learning of Data Literacy Skills. In *Proceedings of the 2022 ACM Conference on International Computing Education Research (2022)*, 2. <https://doi.org/10.1145/3501709.3544281>
- [11] Tamara Munzner. 2014. *Visualization analysis and design*. CRC press.
- [12] Oluwakemi Ola and Kamran Sedig. 2016. Beyond simple charts: Design of visualizations for big health data. *Online Journal of Public Health Informatics* 8 (12 2016). <https://doi.org/10.5210/ojphi.v8i3.7100>
- [13] Srishti Palani, Zijian Ding, Austin Nguyen, Andrew Chuang, Stephen MacNeil, and Steven P. Dow. 2021. CoNotate: Suggesting Queries Based on Notes Promotes Knowledge Discovery. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (Yokohama, Japan) (CHI '21)*. Association for Computing Machinery, New York, NY, USA, Article 726, 14 pages.
- [14] Evan M. Peck, Sofia E. Ayuso, and Omar El-Etr. 2019. Data is Personal: Attitudes and Perceptions of Data Visualization in Rural Pennsylvania. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (Glasgow, Scotland UK) (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–12.
- [15] Ather Sharif, Olivia H. Wang, Alida T. Muongchan, Katharina Reinecke, and Jacob O. Wobbrock. 2022. VoxLens: Making Online Data Visualizations Accessible with an Interactive JavaScript Plug-In. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (New Orleans, LA, USA) (CHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 478, 19 pages.
- [16] Simon Wakeling, Paul D. Clough, Jacqueline F Wyper, and Alexander Balmain. 2015. Graph literacy and business intelligence: Investigating user understanding of dashboard data visualizations.